

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer with 3+ years of experience delivering low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company. Proven track record of collaborating with multinational clients to deploy AI-driven vision solutions in fast-paced dynamic environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes.

WORK EXPERIENCE

GrowthStage • Remote

05/2026 - Present

Recruiter

- Evaluated computer vision and SE candidates on their technical depth across SE practices, model deployment, and MLOps practices, maintaining direct exposure to current production engineering standards.

Taco Bell • Bristol, England

09/2024 - Present

Service Champion / Team Trainer

- Progressed to Team Trainer in a fast-paced customer-facing environment, onboarding new staff and handling customer escalations.

Esbaa • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception in industrial environments, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Developed multimodal data ingestion pipelines processing 0.3 M data points and multiple video streams daily, enhancing data accessibility and analysis efficiency using Airflow and Pandas.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging and structured logging, reducing live video pipeline crashes by 33%.
- Led peer code reviews and introduced pytest and unittest coverage with linting into CI/CD validation workflows for edge inference modules, improving release consistency across hardware targets.
- Integrated AI modules into production systems, collaborating with cross-functional engineering and client teams to deliver fault-tolerant edge deployments 500 mile away.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.
- Designed adversarial training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Partnered with multinational stakeholders, from project managers to rig operators, turning operational problems into deployable data science solutions and communicating results clearly.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.
- Built an activity classification workstream over 91 sensor channels, using **gradient boosting**, **random forests**, and **MultiRocket** for multivariate time-series classification and forecasting.

Lead Software Engineer - AI • Full-time

- Led a 4-engineer team to build a multi-stage detection and helmet-colour classification pipeline for industrial safety and inspection in TensorFlow, reducing false positives by 15% for Austin Industries and John Deere.
- Built low-latency pipelines in C++17 with OpenVINO on edge hw, raising throughput from 5 to 15 FPS, and tuned camera and ISP settings on RGB and infrared streams to hold detection reliable across 24/7 operation.
- Defined and implemented the model optimisation workflow (QAT, pruning, fusion, FP16/INT8) to cut model size by 50% while meeting accuracy, latency and memory targets.
- Automated labelling and dataset versioning, cutting manual labelling effort by 80% and reducing model rollout time from 72 hours to under 24.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using Python and OpenCV.
- Ran systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Mentored junior engineers and collaborated with hardware and QA teams to integrate perception modules into deployed systems with improved delivery consistency.
- Diagnosed recurring detection failures across live camera feeds using confusion-matrix failure clustering, prioritising dataset fixes that improved product robustness.
- Deployed and maintained containerised ML inference microservices on Microsoft Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost per inference across CI/CD and RBAC.
- Designed the PostgreSQL schema that routed real-time video incidents into downstream analytics, supporting worker heat maps, geofencing and client-facing incident feeds.

PROJECTS**DeepStream RTSP Pipeline**

06/2026

- Built a multi-stream NVIDIA DeepStream 9.0 / GStreamer RTSP video analytics pipeline running YOLO detection with NvDCF tracking, metadata extraction, anonymisation and re-streamed output across 3 concurrent camera feeds.
- Optimised a YOLO26n TensorRT FP16 detector on 6GB Turing hardware using dynamic-batch engine conversion and p99 latency profiling, sustaining 29.7 FPS per stream live and 131.3 FPS unthrottled, validated FP16 against FP32 for accuracy.
- Implemented a C++/CUDA TensorRT plugin (IPluginV2DynamicExt) moving post-processing into the inference engine.
- Hardened with GPU smoke tests, model-promotion gates, structured JSON logging and per-camera health metrics covering liveness, dropped frames and reconnect events.

EDUCATION**Master of Science in Robotics - Merit**

University of Bristol • Bristol, England

09/2023 - 09/2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.

SKILLS

- **Cloud & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, containerised microservices, Docker, DVC (Data Version Control), Health checks, Intel VPUs, Microsoft Azure, MLflow, model-promotion gates, observability, Production monitoring, REST APIs, Structured logging

- **Programming & Tooling:** Bash, C++17, CMake, colcon, CUDA C++, Django, FastAPI, Git, Intel VTune, linting, Linux, NumPy, OpenCV, Pandas, PG vector, PIL, Point Cloud Library (PCL), PostgreSQL, pytest, Python, PyTorch, Qt, Scikit-image, Scikit-Learn, TensorFlow, unittest
- **Edge Inference and Optimisation:** Bayesian Hyperparameter Optimisation, Convolutional Neural Networks, custom TensorRT plugins, dynamic batching, FP16/INT8 deployment, latency/throughput profiling, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, Quantisation-Aware Training, Structured pruning, TensorRT, VRAM profiling, YOLO
- **AI-Assisted Development:** ChatGPT Codex, Claude Code
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, GRAD-CAM, Latency benchmarking, Synthetic data generation, TIDE (Toolbox for Identifying Detection Errors)
- **Experimentation and Reliability:** ablation studies, benchmarking, failure-mode analysis, hypothesis testing, post-deployment validation, production monitoring, regression checks
- **Python and ML Engineering:** Apache Airflow, computer vision, DVC, FastAPI, fine-tuning, gradient boosting, LangChain, LangGraph, LangSmith, LoRA, MLflow, MLOps, MultiRocket, NumPy, object detection, Optuna, Pandas, PyDantic, Python, PyTorch, random forests, scikit-learn, Snowflake, time-series classification
- **Time Series Modelling:** gradient boosting, MultiRocket, random forests, Time-series forecasting